

AADITYA SANIL

650-924-5519 | asanil@berkeley.edu | linkedin.com/in/aaditya-sanil | Aaditya Portfolio

EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. Computer Science

Expected Graduation: May 2027

- GPA: **3.70** | Coursework: **Data Structures**, Principles of Data Science, **Optimization Models in Engineering**
- *In Progress*: **Computer Architecture**, **Efficient Algorithms**, **Probability and Random Processes**

EXPERIENCE

Martin Computational Speciation Lab

June 2025 – Present

Undergraduate Research Assistant

Berkeley, CA

- **Refactored** a Julia simulation backend into a **scalable K-buffer architecture** supporting **dynamic population modeling** (100,000+ entities/run) and **distributed execution** across 23 worker processes.
- **Built reproducible data pipelines** exporting 26 structured telemetry fields/run for **downstream statistical analysis**, trend evaluation, and **automated regression validation**.
- **Implemented** defensive sampling, **parameter validation**, and logging safeguards to improve **environment consistency**.

Imagine Games Network (IGN)

Aug. 2025 – Dec. 2025

Data Scientist Intern

Los Angeles, CA

- **Developed training data pipelines** for an Imagine AI by analyzing user queries and structuring **high-quality input/output pairs** using **Python**, **Pandas**, and **Scikit-learn**.
- **Built models** to segment user behavior and intent, enabling more precise **content recommendations** and improving **targeted advertising strategies**.
- **Collaborated with cross-functional teams** to translate model insights into **actionable strategies**, improving advertiser targeting based on user interaction data.

DreamWorks Animation

Feb. 2025 – June 2025

Strategy Consultant

Berkeley, CA

- **Synthesized technical research** on **applied machine learning systems**, evaluating **performance**, **scalability**, and **integration trade-offs** for production pipelines.
- **Designed and presented data-backed recommendations** supporting the **strategic adoption of ML technologies** to optimize **animation rendering workflows**.
- **Translated complex AI/ML models** into **actionable operational strategies**, directly influencing **technical roadmap decisions**.

PROJECTS

Provably Robust Deep Classifiers | Python, PyTorch, NumPy, CVXPY | Github

Jan. 2026 – Present

- Implemented the **Fast Gradient Sign Method (FGSM)** to generate adversarial examples on **MNIST**, demonstrating classifier vulnerability to ℓ_∞ -**bounded input perturbations**.
- Derived **convex relaxations** of ReLU constraints using **Lagrangian duality** and Fenchel conjugates to construct a **dual robustness certificate network**.
- Trained a **provably robust neural network** by minimizing a **worst-case adversarial loss upper bound**, improving **certified robustness** over standard cross-entropy training.

ProteinLLMV1 — Protein Language Model | nanoGPT, React, FastAPI | Github

Dec. 2025 – Present

- Engineered a **custom protein language model** utilizing the **nanoGPT** architecture (12 layers, 12 attention heads), pre-trained via **Masked Language Modeling (MLM)** on UniProt sequences.
- Developed a full-stack web application integrating **ESMFold** to dynamically generate and visualize atomic-level 3D protein structures with confidence scoring via a **React frontend** and **FastAPI backend**.

BMRC Logistics System | Full Stack Web Application, Database | Live Demo

Nov. 2025 – Present

- Engineered a **scalable inventory system** utilizing **relational database design**, structured queries, and **backend validation logic** to ensure operational data integrity.
- Designed **RESTful services** and implemented **CRUD operations** to support **real-time tracking**, dynamic search filtering, and operational reporting for end-users.
- **Demo Test Login**: Test@gmail.com | Password: test123

TECHNICAL SKILLS

Languages: Python, Java, Julia, SQL (PostgreSQL, SQLite), R, C, JavaScript, HTML/CSS, Scheme

Frameworks & APIs: Flask, FastAPI, React, Node.js, REST API Design, JUnit, WordPress, Material-UI

Data & ML: PyTorch, scikit-learn, pandas, NumPy, Matplotlib, joblib, Polars, DuckDB